

Unit 5 Cyber Laws, Digital Forensics & Emerging Trends

Cyber laws and ethics help to:

- Protect user privacy
- Prevent cybercrime
- Ensure safe digital transactions
- Build trust in technology
- Guide professionals to behave responsibly

Cyber Law:

Cyber Law refers to laws that regulate activities in cyberspace such as:

- Internet usage
- Data protection
- Online transactions
- Cybercrime control
- Digital signatures

Example: If someone hacks a bank system and steals money, cyber law defines:

- It is a crime
- What punishment applies
- How investigation should be done

Cyber laws vary by country but share similar goals.

Cyber Ethics:

Cyber Ethics refers to moral principles that guide how we use technology responsibly.

Example:

- Downloading pirated movies → illegal and unethical
- Sharing someone's private photo without permission → unethical
- Respecting others online → ethical behavior

Even if something is not strictly illegal, it can still be unethical.

Principles of Cyber Ethics

1. Respect for Privacy:

Users personal data must be protected.

2. Respect for Intellectual Property:

Software, code, and digital content belong to their creators.

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- No software piracy
- No plagiarism in coding or assignments
- 3. Honesty and Integrity:**
Professionals must be truthful in their work.
 - No fake reporting of results
 - No manipulated data
- 4. Accountability:**
Every action in a system must have responsibility.
 - If a system fails, engineers must take responsibility and fix it
- 5. Fair Use of Technology:**
Technology should not be used to harm others.
 - No hacking systems
 - No spreading malware or misinformation

Professional Responsibilities:

Professional responsibilities are duties that engineers and IT professionals must follow in their workplace.

These responsibilities ensure:

- Safe systems
- Reliable software
- Ethical decision-making
- Public trust in technology

Professional Responsibilities of IT Engineers

1. Ensure security of Systems
2. Maintain Confidentiality
3. Follow Legal Standards
4. Write Secure and Quality code
5. Professional Competence
6. Responsible Communication
7. Avoid Conflicts of Interest

Importance of Cyber Ethics and Professional Responsibilities in IT Industry

In the IT industry, technology is not just used for convenience—it directly impacts people's privacy, finances, communication, healthcare, education, and even national security. Because of this wide influence, cyber ethics and professional responsibilities are extremely important for every IT professional, developer, and cybersecurity expert.

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1. Protecting User Privacy and Data
2. Building Trust in Technology
3. Preventing Cybercrime and misuse of Technology
4. Ensuring Legal Compliance
5. Maintaining Professional Integrity
6. Prompting Secure Software Development
7. Accountability and Responsibility
8. Ethical use of Emerging Technologies
9. Protecting Intellectual Property Rights
10. Enhancing Organizational Reputation

Nepal IT Policy 2015

The Nepal IT Policy 2015 (also called the National Information and Communication Technology Policy 2015) is a government framework that guides the development and use of Information and Communication Technology (ICT) in Nepal.

It was introduced by the Government of Nepal to transform the country into a digital, knowledge-based society.

Main Objectives:

The policy aims to:

- Develop ICT infrastructure in Nepal
- Promote e-governance (online government services)
- Increase digital literacy among citizens
- Support IT industry and startups
- Encourage innovation and research
- Expand internet access in rural areas
- Strengthen cybersecurity and data protection

Areas Covered in the Policy

1. ICT in Education

- Digital learning in schools and universities
- Smart classrooms and e-learning platforms
- ICT skill development for students

2. E-Governance

- Government services through digital platforms
- Online citizenship, tax, and land records
- Transparent public service delivery

3. ICT Industry Development

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- Encouraging IT companies and startups
- Promoting software exports
- Creating IT parks and innovation centers
- 4. Telecommunications Infrastructure**
 - Expanding broadband internet nationwide
 - Improving mobile and internet connectivity
 - Supporting rural connectivity
- 5. Cybersecurity and Data Protection**
 - Strengthening protection against cyber threats
 - Ensuring secure digital transactions
 - Promoting safe use of ICT
- 6. E-Commerce and Digital Economy**
 - Promoting online business platforms
 - Supporting digital payments
 - Encouraging online trade and services
- 7. ICT in Health, Agriculture, and Tourism**
 - Telemedicine services in rural areas
 - Smart farming using ICT tools
 - Digital tourism promotion platforms

Cyber Crimes and Legal Procedures

In today's digital world, almost everything is connected to the internet—banking, education, communication, shopping, and government services. While this brings convenience, it also creates opportunities for misuse.

Cybercrime refers to illegal activities carried out using computers, networks, or digital devices.

To control these crimes, every country follows legal procedures under cyber laws to investigate, punish, and prevent cyber offenses.

Cybercrime: It refers to any crime that involves a computer and network, where the computers may or may not have played an instrumental part in the commission of a crime.

Types of Cyber Crimes:

Spam: Spam, or the unsolicited sending out of junk e-mails for commercial purposes, which is unlawful. New anti-spam laws are being passed in various countries which will hopefully limit the use of unsolicited electronic communications.

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Hacking:

- Gaining access to systems without permission
- Example: Breaking into a website admin panel

Identity Theft

- Stealing personal information like passwords or bank details
- Example: Using someone's bank account credentials

Phishing Attacks

- Fake emails or websites used to steal sensitive data
- Example: Fake banking login page

Online Financial Fraud

- ATM fraud, online shopping scams, fake investment schemes

Cyber Stalking

- Harassing or threatening someone online repeatedly

Cyber Bullying

- Using social media to insult, threaten, or defame others

Denial of Service (DoS/DDoS)

- Overloading a system to make it unavailable

Malware Attacks

- Virus, worm, trojan, ransomware
- Designed to damage or steal data

Data Theft

- Stealing confidential company or government data

Software Piracy

- Illegal copying and distribution of software

Intellectual Property Rights (IPR) in Cyberspace

In the digital world, almost everything is created, shared, and stored in electronic form—software, music, videos, images, code, and documents. These creations are valuable intellectual work of individuals or organizations.

To protect these creations, we use Intellectual Property Rights.

Intellectual property in cyberspace includes:

- Software programs and source code
- Websites and mobile applications

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- Digital images, videos, and music
- Databases and online content
- Algorithms and AI models

Types of Intellectual Property Rights

1. Copyright

Copyright protects original digital content.

Protected items:

- Software code
- Books and articles (digital format)
- Music and videos
- Website content

Example: A developer writes a mobile app. No one can copy its source code without permission.

2. Trademark

A trademark protects brand identity.

Protected items:

- Logos
- Brand names
- Slogans
- Website identity

Example:

The logo of a company like Google or Amazon cannot be copied or misused.

3. Patent

A patent protects inventions and technical innovations.

In cyberspace:

- New algorithms
- AI models
- Software innovations (in some countries)

Example:

A unique encryption technique developed by a company can be patented.

4. Trade Secrets

Trade secrets are confidential business information.

Examples:

- Source code of a proprietary software

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- Business algorithms (like search ranking systems)
- Database structures

Privacy and Data Protection laws (GDPR overview)

In today's digital world, almost every online activity involves personal data—such as name, phone number, email, location, photos, banking details, and browsing behavior.

Because data is so valuable, it can be misused for:

- identity theft
- financial fraud
- surveillance
- unauthorized marketing
- data breaches

To protect individuals, governments have created data protection laws. One of the most important and strict laws in the world is the GDPR of the European Union (EU).

GDPR

GDPR (General Data Protection Regulation) is a legal framework introduced by the European Union in May 2018 to protect personal data and privacy of individuals.

Even companies outside Europe must follow GDPR if they handle EU citizens' data.

Objective of GDPR

GDPR was designed to:

- Protect personal data and privacy
- Give control to users over their data
- Ensure transparency in data usage
- Prevent data misuse and breaches
- Standardize data protection laws across Europe

Principle of GDPR

1. **Lawfulness, Fairness, and Transparency**
2. **Purpose Limitation**
3. **Data Minimization**
4. **Accuracy**
5. **Storage Limitation**
6. **Integrity and Confidentiality**
7. **Accountability**

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8. Rights of Individuals under GDPR

GDPR gives strong rights to users:

a. Right to Access

Users can ask what data is stored about them.

b. Right to Rectification

Users can correct incorrect data.

c. Right to Erasure (Right to be Forgotten)

Users can request deletion of their data.

d. Right to Data Portability

Users can transfer their data to another service.

e. Right to Restrict Processing

Users can limit how their data is used.

f. Right to Object

Users can refuse data processing for marketing purposes.

g. Responsibilities of Organizations

Companies that collect data must:

- Get user consent before collecting data
- Clearly explain how data will be used
- Secure data with encryption and security controls
- Report data breaches within 72 hours
- Appoint Data Protection Officers (DPOs) in some cases

h. Data Breach Notification Rule

If a data breach occurs:

- Organization must notify authorities within 72 hours
- Affected users must be informed if risk is high

i. GDPR in Cybersecurity Context

GDPR is closely linked with cybersecurity:

- Encourages strong encryption
- Promotes secure system design
- Requires regular security audits
- Reduces chances of data leaks

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5.2 Digital Forensics

In today's digital world, almost every activity leaves a digital footprint—emails, messages, browsing history, social media activity, transaction logs, and even deleted files (which can often still be recovered).

When a cybercrime or digital incident happens, investigators need scientific methods to find evidence from digital devices.

Digital Forensics is the process of identifying, collecting, preserving, analyzing, and presenting digital evidence in a way that is legally acceptable in court.

Important of Digital Forensics

Digital forensics is important because:

- Cybercrimes are increasing
- Evidence exists in digital form
- Criminals hide or delete data
- Courts require valid proof
- Helps identify attackers

Objective of Digital Forensics

The main goals are:

- Identify cybercrime activities
- Recover deleted or hidden data
- Preserve digital evidence safely
- Analyze system logs and files
- Present evidence in court

Digital Evidence:

Any information stored or transmitted in digital form that can be used in an investigation.

Examples:

- Emails and attachments
- Chat messages (WhatsApp, Messenger)
- System logs
- Images, videos, audio files
- Browsing history
- Deleted files (recoverable data)

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Characteristics of Digital Evidence

Digital evidence must be:

1. Admissible

Accepted in court as legal proof.

2. Authentic

Must be real and not tampered with.

3. Reliable

Accurately represents original data.

4. Complete

Should provide full context of the incident.

Process of Digital Forensics

Digital forensics follows a structural process.

Step 1: Identification

- Identify potential sources of evidence
- Example: laptops, mobile phones, servers

Step 2: Preservation

- Protect evidence from modification
- Create backup copies (forensic imaging)
- Maintain chain of custody

Step 3: Collection

- Gather data from devices
- Extract files, logs, and metadata

Step 4: Examination

- Filter irrelevant data
- Recover deleted files
- Analyze hidden information

Step 5: Analysis

- Interpret data to find:
 - Who committed the crime
 - When it happened
 - How it happened

Step 6: Presentation

- Present findings in court
- Prepare reports understandable to non-technical people

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Types of Digital Forensics

1. Computer Forensics

Investigation of computers and storage devices

2. Network Forensics

Analysis of network traffic and logs

3. Mobile Forensics

Investigation of smartphones and tablets

4. Cloud Forensics

Investigation of cloud storage and services

5. Database Forensics

Analysis of database transactions and logs

Evidence Collection and Preservation in Digital Forensics

In cybercrime investigations, the most important part is digital evidence. Without proper evidence, it is impossible to prove what happened, who did it, and how it was done.

However, digital evidence is very fragile—it can be easily modified, deleted, or corrupted. That is why evidence collection and preservation is the most critical phase in Digital Forensics

Evidence Collection

Evidence collection is the process of gathering digital data from devices, networks, or storage systems in a systematic and legal manner.

Evidence Preservation

Evidence preservation means protecting collected evidence from alteration, damage, or loss so that it remains valid for investigation and court use.

Important of Evidence Collection and Preservation

- Cyber evidence can be easily deleted or modified
- Courts require authentic and unaltered data
- Small mistakes can destroy legal validity
- Ensures justice in cybercrime cases
- Helps maintain trust in investigation process

Types of Digital Evidence

1. Computer Devices

- Hard disks
- Laptops
- Servers

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2. Mobile Devices

- Smartphones
- Tablets
- SIM data

3. Network Data

- IP logs
- Firewall logs
- Router logs

4. Online Sources

- Emails
- Social media data
- Cloud storage

Network and System Forensics

In cybersecurity investigations, evidence is not always stored in one place. Sometimes, attackers leave traces:

- in network traffic
- in system logs
- in running processes
- in memory (RAM)

To analyze these traces, we use two important branches of digital forensics:

- Network Forensics
- System Forensics

Network Forensics

Network forensics is the process of monitoring, capturing, and analyzing network traffic to detect and investigate cyber incidents.

Objectives of Network Forensics

- Identify attackers and their location
- Detect malicious activity
- Analyze data breaches
- Reconstruct network attacks
- Support legal investigations

Sources of Network Evidence

Network forensic experts analyze:

- Packet data (TCP/IP packets)

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- Router logs
- Firewall logs
- DNS logs
- Proxy server logs
- IDS/IPS alerts

System Forensics

System forensics is the process of analyzing computer systems (hardware and software) to find evidence of cybercrime.

Objective of System Forensics

- Identify unauthorized access
- Recover deleted files
- Analyze malware behavior
- Examine system logs
- Detect system misuse

Incident Response Cycle:

In cybersecurity, **security incidents** are unavoidable. These may include:

- Hacking attempts
- Malware infections
- Data breaches
- Denial of Service (DoS/DDoS) attacks
- Insider threats

When such events occur, organizations must respond quickly and systematically to reduce damage. This structured approach is called Incident Response Lifecycle.

Incident Response: Incident Response is the process of detecting, handling, and recovering from cybersecurity incidents in a controlled and organized manner.

Objective of Incident Response

- Minimize damage and downtime
- Reduce financial loss
- Protect sensitive data
- Restore normal operations
- Identify the attacker and method
- Prevent future incidents

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Incident Response Lifecycle (Core Phases)

Phase 1: Preparation

Phase 2: Identification (Detection)

Phase 3: Containment

Phase 4: Eradication

Phase 5: Recovery

Phase 6: Lessons Learned

5.3 Emerging Trends

Artificial Intelligence in Cyber Security

Blockchain and Security

Zero Trust Architecture

Future Challenges in cyber security